

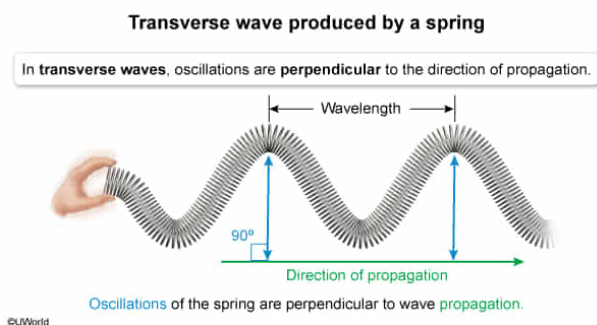
A wave is sent along a spring that is horizontally oriented, as shown above. If the wave moves toward the right, what kind of wave is shown, and which statement explains why?

- ☐ A. The wave is longitudinal because the direction of motion of the wave is toward the right. [5%]
- ☐ B. The wave is longitudinal because the spring oscillates in a direction that is parallel to the wave's direction of motion. [32%]
- ☐ C. The wave is transverse because the direction of motion of the wave is toward the right. [4%]
- ✓ ☒ D. The wave is transverse because the spring oscillates in a direction that is perpendicular to the wave's direction of motion. [58%]

Correct

58% answered correctly

Explanation:



A mechanical **wave** is a motion created by a disturbance in a medium. Mechanical waves can propagate through a medium in two different wave forms, **transverse and longitudinal**. **Transverse waves** displace the components of the medium **perpendicular** to the direction of the wave's propagation. However, **longitudinal waves** displace the medium in a direction **parallel** to the direction of the wave's propagation.

An example of a transverse wave is a water wave in the ocean, where the water rises and falls vertically relative to the ocean surface, but the wave travels horizontally parallel to the surface. On the other hand, **sound waves** are an example of longitudinal waves because they create disturbances in the air, causing air molecules

to oscillate back and forth parallel to the direction the sound wave travels through the air.

In this question, the spring (the medium) is displaced in a direction perpendicular to the direction of the wave's motion. Therefore, the wave produced by the spring is a transverse wave.

(Choices A and C) The direction of wave motion alone does not define the type of wave. The direction of wave oscillations is also needed.

(Choice B) This answer choice correctly defines a longitudinal wave. However, the wave shown is a transverse wave because its oscillations are perpendicular to the direction of propagation.

Educational objective:

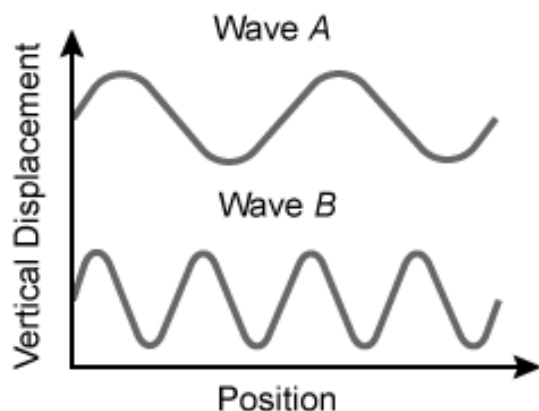
Transverse waves displace a medium perpendicular to the direction of the wave's propagation. Longitudinal waves displace a medium parallel to the direction of the wave's propagation.

Time spent:0 sec

Subject:
Physics

QID:403946

System:
Light and Sound



Vertical displacement as a function of position for two waves passing horizontally through the same medium is shown in the figure above. Which of the following is true about the waves?

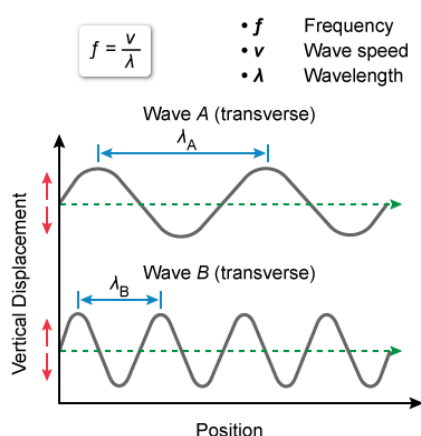
- ☐ A. The waves are transverse; wave *A* has a higher frequency than wave *B*. [2%]
- ✓ ☒ B. The waves are transverse; wave *A* has a lower frequency than wave *B*. [76%]
- ☐ C. The waves are longitudinal; wave *A* has a lower frequency than wave *B*. [17%]
- ☐ D. Wave *A* is longitudinal, but wave *B* is transverse; wave *A* has a lower frequency than wave *B*. [3%]

Correct

75% answered correctly

Explanation:

Wave type and frequency from a graph of displacement versus position



Transverse waves have **displacements** perpendicular to the **direction of motion**. A longer wavelength ($\lambda_A > \lambda_B$) gives a lower frequency ($f_A < f_B$).

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A **wave** is a perturbation that propagates from one point to another. The speed of

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wave propagation v is the product of the wavelength λ and the frequency f :

$$v = \lambda f$$

Waves can be classified into two types: **longitudinal or transverse**. For longitudinal waves, the direction of the oscillatory perturbation is parallel to the direction of wave propagation. In contrast, transverse waves are characterized by perturbations **perpendicular** to the direction of wave motion.

In this question, the vertical displacement from equilibrium of waves is measured as they pass by in the horizontal direction. Consequently, the perturbations are perpendicular to the direction of propagation and the waves are both **transverse**. The frequency of each wave can be determined by rearranging the wave speed equation:

$$f = \frac{v}{\lambda}$$

Because both waves are traveling through the **same medium**, the speed of each wave is the same. As shown, frequency is **inversely proportional** to wavelength. The graph shows that wave *A* has a longer wavelength than wave *B*, and so it must have a lower frequency. Therefore, both waves are transverse and wave *A* has a lower frequency than wave *B*.

(Choice A) Both waves are transverse, but wave *A* has a lower frequency than wave *B* because wave *A* has a longer wavelength.

(Choices C and D) Wave *A* does have a lower frequency than wave *B*, but both waves are transverse.

Educational objective:

Transverse waves perturb the medium in a direction perpendicular to the direction of their motion. The frequency of a wave is inversely proportional to its wavelength.

Time spent:0 sec

Subject:
Physics

QID:403947

System:
Light and Sound